Requirements Coverage Report

COMP3211 Software Engineering | Fall 2026

Requirements Coverage Report

Status: Stage 5 verification result

The table maps each mandatory user story to executable tests. The latest run used Python 3.12.7 on 2026-10-28 and completed 31 tests with 0 failures and 0 errors.

User story	Verification tests	Result
US1	`test_pir.py::PIRConstructionTests.test_all_record_types_store_required_fields`; `test_cli.py::ControllerTests.test_controller_supports_create_update_search_print_and_delete`	PASS
US2	`test_pir.py::PIRConstructionTests.test_all_record_types_store_required_fields`; `test_pir.py::PIRConstructionTests.test_required_values_and_identifiers_are_validated`; `test_main.py::MainLoopTests.test_run_processes_commands_and_continues_after_syntax_errors`	PASS
US3	`test_pir.py::PIRConstructionTests.test_all_record_types_store_required_fields`; `test_pir.py::PIRConstructionTests.test_required_values_and_identifiers_are_validated`; `test_pir.py::PIRUpdateTests.test_updates_change_fields_without_changing_identity_or_type`	PASS
US4	`test_pir.py::PIRConstructionTests.test_all_record_types_store_required_fields`; `test_pir.py::PIRConstructionTests.test_required_values_and_identifiers_are_validated`; `test_search.py::SearchTests.test_time_search_skips_records_without_the_requested_field`	PASS
US5	`test_pir.py::PIRConstructionTests.test_all_record_types_store_required_fields`; `test_pir.py::PIRConstructionTests.test_required_values_and_identifiers_are_validated`; `test_pir.py::PIRUpdateTests.test_updates_change_fields_without_changing_identity_or_type`	PASS
US6	`test_pir.py::PIRUpdateTests.test_updates_change_fields_without_changing_identity_or_type`; `test_pir.py::PIRUpdateTests.test_invalid_multi_field_update_is_atomic`; `test_cli.py::ControllerTests.test_controller_reports_errors_without_corrupting_repository`	PASS
US7	`test_search.py::SearchTests.test_type_and_case_insensitive_text_search`; `test_search.py::SearchTests.test_time_search_skips_records_without_the_requested_field`; `test_search.py::SearchTests.test_boolean_precedence_and_parentheses`; `test_search.py::SearchTests.test_invalid_criteria_are_rejected`	PASS
US8	`test_repository.py::RepositoryTests.test_replace_sorts_records_and_resets_next_identifier`; `test_cli.py::ControllerTests.test_controller_supports_create_update_search_print_and_delete`; `test_cli.py::ViewTests.test_view_renders_each_result_kind`	PASS
US9	`test_repository.py::RepositoryTests.test_get_update_delete_and_missing_ids`; `test_cli.py::ControllerTests.test_controller_supports_create_update_search_print_and_delete`	PASS

User story	Verification tests	Result
US10	`test_persistence.py::PersistenceTests.test_round_trip_preserves_all_record_types_and_ids`; `test_persistence.py::PersistenceTests.test_save_validates_path_and_record_values`; `test_cli.py::ControllerTests.test_controller_save_load_and_failed_load_are_safe`	PASS
US11	`test_persistence.py::PersistenceTests.test_load_rejects_missing_malformed_incompatible_and_duplicate_files`; `test_cli.py::ControllerTests.test_controller_save_load_and_failed_load_are_safe`	PASS

Additional non-functional verification

- **NFR-01 / NFR-02:** tests import the model and CLI packages directly using the Python standard library; no network or third-party runtime dependency is used.
- **NFR-03 / NFR-05:** parser, controller, view, and main-loop tests verify human-readable error results and continued command processing.
- **NFR-04:** persistence round-trip tests compare every serialized record and identifier.
- **NFR-06:** repository tests verify stable ID ordering for print and search results.
- **NFR-08:** successful and validation-failure paths are covered for public model operations.
- **NFR-09:** tests use workspace-local files and do not contact a network service.